

Matrix Operations in Practise

1. Why Do We Need Classification Metrics?

After training a classification model, the next question is:

"How good is the model?"

To answer this, we use **evaluation metrics**.

For classification problems, the model predicts categories such as:

- Spam / Not Spam
- Fraud / Not Fraud
- Disease / No Disease
- Pass / Fail

The video starts with the simplest metric:

Accuracy

Accuracy measures the percentage of correct predictions.

Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Example

Suppose:

- Total observations = 100
- Correct predictions = 90

Then:

Accuracy = $90/100 = 90\%$

The model is correct 90% of the time.

2. Accuracy in Scikit-Learn

The video demonstrates how to calculate accuracy using Python's **Scikit-Learn** library.

Using:

```
accuracy_score(y_true, y_pred)
```

This function compares actual labels with predicted labels and returns the accuracy score.

3. Accuracy in Multi-Class Classification

Accuracy is not limited to binary classification.

Example:

Classes:

- Cat
- Dog
- Horse

If a model predicts the correct class for 85 out of 100 examples:

Accuracy = 85%

The same formula applies regardless of the number of classes.

4. How Much Accuracy Is Considered Good?

A very important point discussed in the video:

There is **no universal rule**.

For some problems:

- 70% may be excellent.

For others:

- 99% may still be poor.

The answer depends on:

Dataset Complexity

Harder problems naturally achieve lower accuracy.

Business Requirements

Different applications tolerate different mistakes.

Examples:

- Movie recommendation → 80% may be acceptable.
- Cancer detection → 80% is dangerous.

Thus, accuracy must always be interpreted in context.

5. The Biggest Problem with Accuracy

The video emphasizes that accuracy can be highly misleading.

Consider a dataset:

Class	Count
Negative	990
Positive	10

Suppose a model predicts:

"Negative" for every observation.

Then:

- Correct predictions = 990
- Wrong predictions = 10

Accuracy:

99%

Looks amazing.

But:

The model never detects a positive case.

Therefore, the model is actually useless.

This situation is called an **imbalanced dataset problem**. Accuracy often fails when one class dominates the dataset.

6. Confusion Matrix

To solve the limitations of accuracy, we use the **Confusion Matrix**.

A confusion matrix shows:

- What the model predicted
- What the actual values were

Binary confusion matrix:

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

True Positive (TP)

Actual = Positive

Predicted = Positive

Example:

Patient has disease.

Model predicts disease.

Correct prediction.

True Negative (TN)

Actual = Negative

Predicted = Negative

Example:

Healthy patient.

Model predicts healthy.

Correct prediction.

False Positive (FP)

Actual = Negative

Predicted = Positive

Example:

Healthy patient diagnosed as sick.

Wrong prediction.

False Negative (FN)

Actual = Positive

Predicted = Negative

Example:

Sick patient diagnosed as healthy.

Wrong prediction.

7. Type I Error

Type I Error = False Positive

The model predicts positive when reality is negative.

Examples:

Spam Detection

Legitimate email marked as spam.

Medical Testing

Healthy person diagnosed with disease.

Fraud Detection

Valid transaction marked as fraud.

Relationship:

Type I Error = False Positive (FP)

8. Type II Error

Type II Error = False Negative

The model predicts negative when reality is positive.

Examples:

Medical Diagnosis

Patient actually has cancer.

Model says healthy.

Fraud Detection

Fraudulent transaction passes undetected.

Spam Filtering

Spam email enters inbox.

Relationship:

Type II Error = False Negative (FN)

9. Which Error Is Worse?

The video explains that the answer depends on the application.

Disease Detection

Type II Error is usually worse.

Reason:

Missing a disease can be life-threatening.

Spam Detection

Type I Error is often worse.

Reason:

Important emails may be lost.

Fraud Detection

Type II Error is usually more costly.

Reason:

Fraud goes undetected.

The cost of errors determines which metric should be prioritized.

10. Confusion Matrix for Multi-Class Problems

The video also extends the confusion matrix concept to multi-class classification.

Example classes:

- Cat
- Dog
- Horse

Instead of a 2×2 matrix, we create an N×N matrix.

For 3 classes:

Actual \ Predicted	Cat	Dog	Horse
Cat	✓	✗	✗
Dog	✗	✓	✗
Horse	✗	✗	✓

Diagonal values represent correct predictions.

Off-diagonal values represent mistakes.

This helps identify exactly which classes are being confused.

11. When Accuracy Becomes Misleading

The final major lesson:

Never rely solely on accuracy.

Accuracy becomes unreliable when:

- The dataset is imbalanced.
- The cost of mistakes is unequal.
- Minority class is important.
- Medical or fraud applications are involved.

In such cases, confusion matrix-based metrics provide a better understanding of model behavior.

Dimensionality Reduction Basics

1. What is Dimensionality?

A **dimension** simply means a feature or variable.

Example:

House	Area	Bedroom s	Price
House 1	1000	2	20L

Features:

- Area
- Bedrooms

So this dataset has **2 dimensions**.

If we add:

- Age
- Location Score
- Distance from Market
- Parking Size

then dimensions increase.

In Machine Learning:

- 10 features → 10-dimensional space
- 100 features → 100-dimensional space
- 1000 features → 1000-dimensional space

The video emphasizes that higher dimensions create unexpected mathematical problems.

2. Intuition Through Geometry

The video starts with a geometric explanation.

1-Dimensional Space

A line:

To cover the line, only a few points are needed.

2-Dimensional Space

A square.

Area increases.

More sample points are needed.

3-Dimensional Space

A cube.

Volume increases even more.

Need many more observations.

Higher Dimensions

When dimensions keep increasing:

- Space grows exponentially.
- Data points become extremely sparse.
- Distances become less meaningful.

This phenomenon is called the Curse of Dimensionality.

3. Why Does Sparsity Happen?

The key idea:

As dimensions increase, available data spreads out.

Example:

Suppose:

- 100 data points in 1D

Points are relatively close.

Now use the same 100 points in 100D.

The points become extremely far apart.

Large portions of the feature space remain empty.

This is called **data sparsity**.

4. The Hypercube Example

CampusX explains this using a hypercube.

Imagine a cube with side length = 1.

Volume formula:

1D

Length = 1

Volume = 1

2D

Area:

$\text{Area} = s^2$

For $s = 1$:

Area = 1

3D

Volume:

$$\text{Volume} = s^3$$

For $s = 1$:

$$\text{Volume} = 1$$

The surprising part:

To maintain the same density of data points, the number of required samples grows exponentially with dimensions.

Example:

- 10 points per dimension

Then:

1D:

10 points

2D:

100 points

3D:

1000 points

10D:

$$10^{10} = 10,000,000,000$$

Ten billion points.

This is practically impossible.

5. Distance Becomes Meaningless

Many ML algorithms depend on distance:

- KNN

- K-Means
- Hierarchical Clustering

The video explains that in high dimensions:

Nearest points and farthest points begin to look equally distant.

Example

Imagine points in 2D.

You can clearly identify neighbors.

But in 100 dimensions:

Almost every point is far away.

Difference between:

- nearest neighbor
- farthest neighbor

becomes very small.

Thus distance-based algorithms struggle to identify meaningful patterns.

6. Effect on K-Nearest Neighbors (KNN)

The video particularly discusses KNN.

KNN works by:

1. Calculating distance.
2. Finding nearest neighbors.
3. Predicting based on neighbors.

When dimensions increase:

- All points become far apart.
- Neighbor concept weakens.
- Prediction quality decreases.

Therefore KNN suffers heavily from the Curse of Dimensionality.

7. Overfitting Problem

Another major issue:

Higher dimensions increase model complexity.

The model starts memorizing training data.

Instead of learning patterns, it learns noise.

This causes:

Training Accuracy

Very high

Test Accuracy

Low

This is called **overfitting**.

The video highlights that adding features is not always beneficial. More features can actually reduce performance.

8. Hughes Phenomenon (Peaking Phenomenon)

An important concept connected to the Curse of Dimensionality.

Initially:

Adding features improves performance.

After a certain point:

Adding more features hurts performance.

Graphically:

Accuracy

^

|

| ^

| / \

| / \

|____/_________> Features

Performance peaks and then drops.

This phenomenon is called the **Hughes Phenomenon**.

9. Why More Features Are Not Always Better

Many beginners think:

More information = Better model

But the video explains:

Extra features often include:

- Noise
- Redundant information
- Irrelevant information

These confuse the model.

As dimensionality grows:

- Learning becomes harder.
- Generalization becomes weaker.

Hence feature quality matters more than feature quantity.

10. How to Handle the Curse of Dimensionality

The video discusses dimensionality reduction techniques.

A. Feature Selection

Keep only important features.

Remove:

- Irrelevant features
- Duplicate features
- Weak predictors

Methods:

- Correlation Analysis
 - Mutual Information
 - Chi-Square Test
-

B. Dimensionality Reduction

Transform many features into fewer features.

Popular methods:

PCA

Principal Component Analysis

Creates new dimensions that preserve maximum variance.

LDA

Linear Discriminant Analysis

Projects data while maximizing class separation.

t-SNE

t-distributed Stochastic Neighbor Embedding

Useful for visualization of high-dimensional data.

11. Real-Life Example

Consider image classification.

A grayscale image:

100 × 100 pixels

Features:

$$100 \times 100 = 10000 \quad 100 \times 100 = 10000 \quad 100 \times 100 = 10000$$

10,000 dimensions.

For color images:

RGB channels:

$$100 \times 100 \times 3 = 30000 \quad 100 \times 100 \times 3 = 30000 \quad 100 \times 100 \times 3 = 30000$$

30,000 dimensions.

Without dimensionality reduction:

- Training becomes difficult.
- Storage increases.
- Computation becomes expensive.

This is why feature extraction is crucial in computer vision.

Stats and its types

Types of Statistics

There are **two main types of statistics**:

1. Descriptive Statistics

Descriptive statistics is used to **summarize and describe data**. It tells us what the data looks like but does not make predictions about a larger population.

Examples

Suppose marks of students are:

70, 75, 80, 85, 90

Descriptive statistics helps calculate:

- Mean (Average)
- Median
- Mode
- Range
- Variance
- Standard Deviation

Common Measures

Mean

$$\bar{x} = \sum x_i / n$$

Variance

$$\text{Variance} = \sum (x_i - \bar{x})^2 / n$$

Standard Deviation

$$SD = \sqrt{\text{Variance}}$$

Tools Used

- Tables
- Histograms
- Pie Charts
- Bar Charts
- Box Plots

Goal

Describe the dataset.

2. Inferential Statistics

Inferential statistics is used to **draw conclusions about a population using sample data**.

Instead of studying the entire population, we study a sample and make predictions.

Example

A university has:

20,000 students

Survey:

100 students

If the sample GPA average is 8.0, inferential statistics helps estimate the GPA of all 20,000 students.

Hypothesis Testing

What are Variables?

In statistics and data science, a **variable** is any characteristic, attribute, or feature that can take different values for different observations.

Examples

- Age of a person
- Height
- Weight
- Salary
- Gender
- City
- Exam Marks

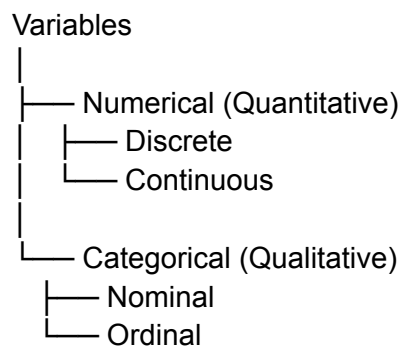
For example, if you collect data from 100 students:

Student	Age
A	18
B	20

Here, **Age** is a variable because its value changes from student to student.

Types of Variables

Krish Naik typically classifies variables into **two major categories**:



1. Numerical (Quantitative) Variables

These variables contain numeric values and support mathematical operations.

A. Discrete Variables

A discrete variable can take **countable values**.

Examples:

- Number of students in a class
- Number of cars in a parking lot
- Number of children in a family

Possible values:

0, 1, 2, 3, 4, ...

You cannot have:

2.5 children

3.7 cars

Example

Family	Children
A	2
B	4
C	1

The number of children is a discrete variable.

B. Continuous Variables

A continuous variable can take **any value within a range**, including decimals.

Examples:

- Height
- Weight
- Temperature
- Salary
- Distance

Example

Perso n	Height (cm)
A	170.5
B	168.2
C	175.8

Height can have infinitely many values, so it is continuous.

2. Categorical (Qualitative) Variables

These variables represent categories or labels rather than numerical quantities.

A. Nominal Variables

Nominal variables have categories with **no natural order**.

Examples:

- Gender
- Religion
- Blood Group
- City
- Country

Example

Perso n	Blood Group
A	A+
B	O+
C	B+

There is no ranking among blood groups.

Hence, Blood Group is a nominal variable.

B. Ordinal Variables

Ordinal variables have categories with a meaningful order or ranking.

Examples:

- Education Level
- Customer Satisfaction
- Movie Ratings

Example

Customer Satisfaction:

Poor < Average < Good < Excellent

The categories are ordered, but the difference between them is not precisely measurable.

Therefore, it is an ordinal variable.

Interview Examples

Example 1

Variable: Age

Answer:

- Numerical
 - Continuous
-

Example 2

Variable: Number of Employees

Answer:

- Numerical
 - Discrete
-

Example 3

Variable: Gender

Answer:

- Categorical
 - Nominal
-

Example 4

Variable: Customer Rating

1 Star

2 Star

3 Star

4 Star

5 Star

Answer:

- Categorical
 - Ordinal
-

Why Are Variable Types Important in Machine Learning?

Different variable types require different preprocessing techniques.

Numerical Variables

Common preprocessing:

- Scaling
- Normalization
- Standardization

Categorical Variables

Common preprocessing:

- Label Encoding
- One-Hot Encoding

For example, a machine learning model cannot directly understand:

Red
Blue
Green

These categories must first be converted into numbers.

Error Matrix and Graphical Intuition

Measure of Dispersion – Variance and Standard Deviation

After learning measures of central tendency (**Mean, Median, Mode**), the next question is:

How spread out is the data around the mean?

This spread is called **dispersion** or **variability**.

Two datasets can have the same mean but very different distributions.

Example

Dataset A:

10, 10, 10, 10, 10

Mean = 10

Dataset B:

1, 5, 10, 15, 19

Mean = 10

Although both have the same meaning, Dataset B is much more spread out.

This spread is measured using **Variance** and **Standard Deviation**.

What is Variance?

Variance measures the average squared distance of data points from the mean.

Population Variance

Population Variance

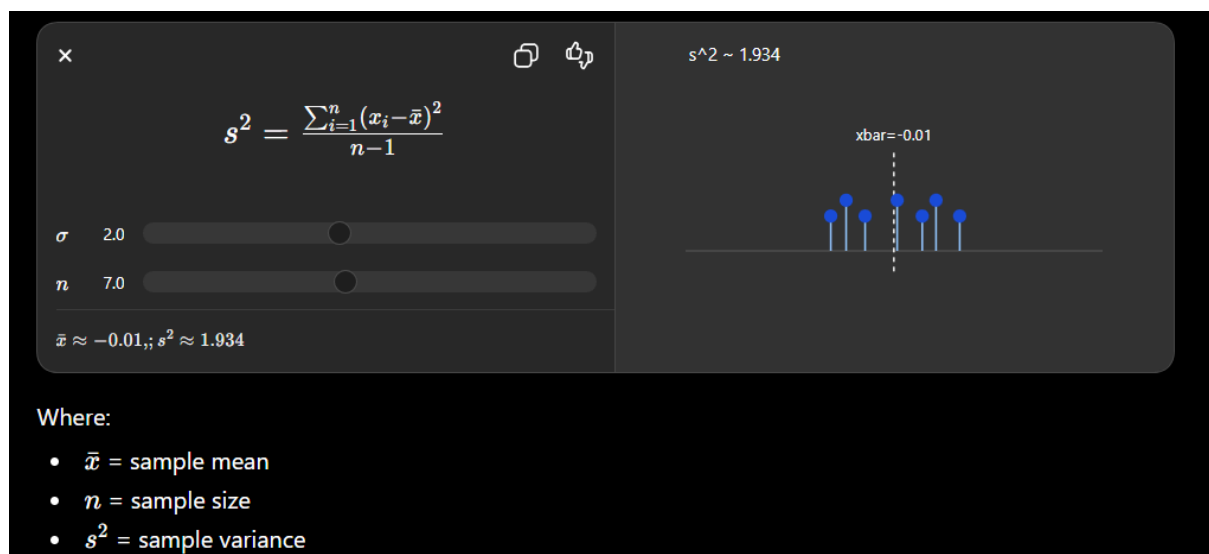
$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Where:

- x_i = each observation
- μ = population mean
- N = population size
- σ^2 = population variance

Sample Variance

When working with a sample instead of the whole population:



The $n-1$ term is called **Bessel's correction**, which gives a better estimate of population variance.

Variance Calculation Example

Consider:

2, 4, 6, 8, 10

Step 1: Calculate Mean

$$\bar{x} = (2+4+6+8+10)/5 = 6$$

Mean = 6

Step 2: Find Deviations from Mean

Value	Deviation
2	-4
4	-2
6	0
8	2
10	4

Notice:

$$-4 + (-2) + 0 + 2 + 4 = 0$$

Positive and negative deviations cancel out.

To solve this, we square them.

Step 3: Square Deviations

Deviation	Squared
-4	16
-2	4
0	0
2	4

4

16

Sum:

$$16 + 4 + 0 + 4 + 16 = 40$$

Step 4: Calculate Variance

Population Variance:

$$\sigma^2 = 40/5 = 8$$

Variance = 8

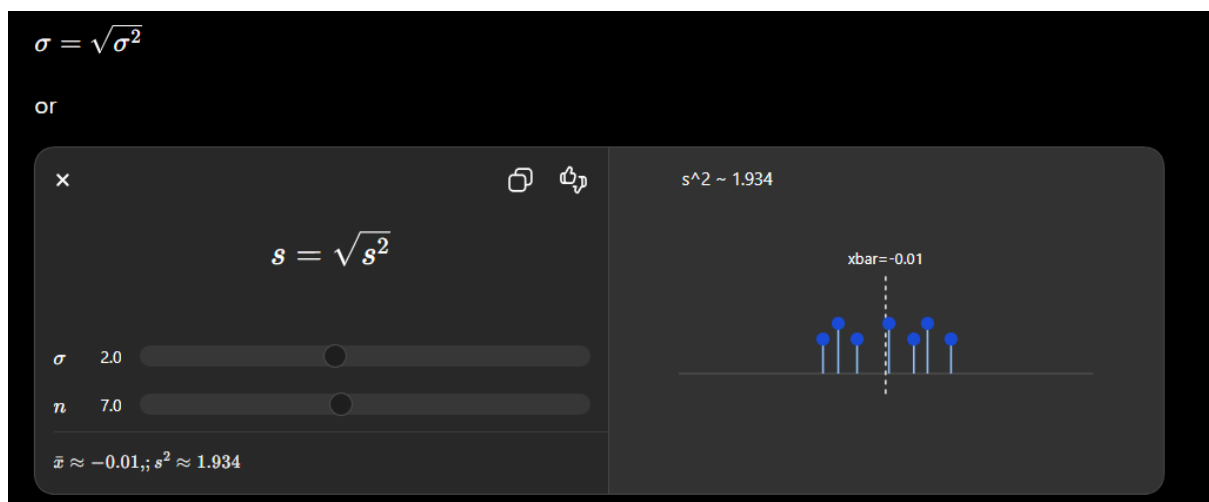
What is Standard Deviation?

One drawback of variance:

- Original data unit might be cm, kg, ₹, etc.
- Variance is in squared units (cm², kg², ₹²).

To bring it back to the original unit, take the square root.

Formula



Example

Variance = 8

Standard Deviation:

$$\sigma = \sqrt{8} = 2.83$$

So the average distance from the mean is approximately **2.83 units**.

Interpretation

Low Standard Deviation

9, 10, 11, 10, 10

- Data points are close to mean.
 - Low variability.
-

High Standard Deviation

1, 5, 10, 15, 19

- Data points are far from mean.
 - High variability.
-

Real-Life Example

Suppose two classes have:

Class	Average Marks
A	70
B	70

Same average.

But:

Class A

68, 69, 70, 71, 72

Students perform similarly.

Small standard deviation.

Class B

20, 40, 70, 90, 130

Huge differences in performance.

Large standard deviation.

Thus, standard deviation gives information that mean alone cannot provide.

Why Standard Deviation is Important in Machine Learning

It is used in:

Data Analysis

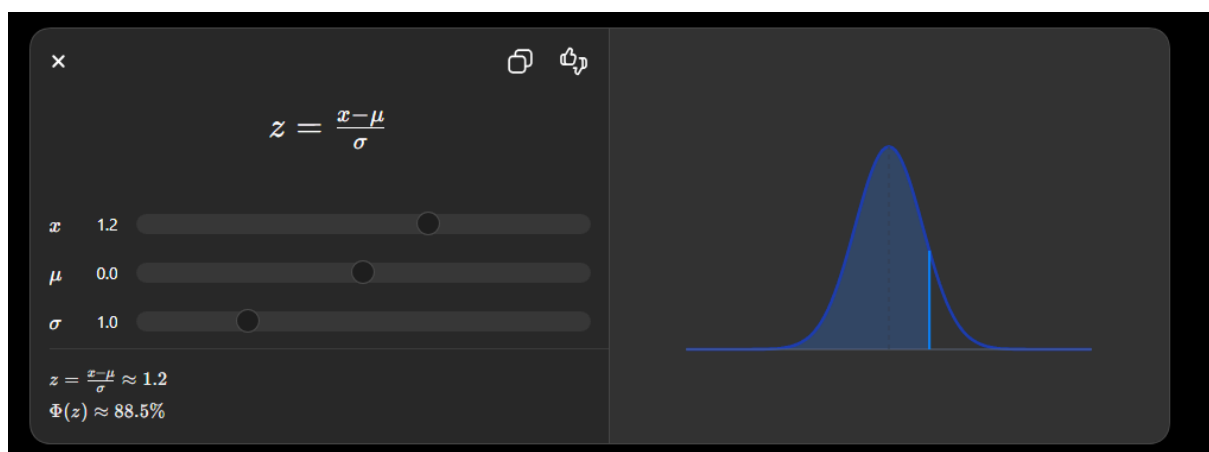
Understanding spread and variability.

Outlier Detection

Values far from the mean are often outliers.

Standardization (Z-Score)

Used before training many ML algorithms.



Probability Distributions

Especially the Normal Distribution.

Variance vs Standard Deviation

Feature	Variance	Standard Deviation
Definition	Average squared deviation	Square root of variance
Unit	Squared unit	Original unit
Interpretation	Less intuitive	More intuitive
Formula Output	σ^2	σ

Interview Question

Why do we square deviations while calculating variance?

Because if we simply add deviations from the mean, positive and negative values cancel each other and the result becomes zero. Squaring makes all values positive and gives a meaningful measure of spread.

Quick Revision

Measure of Dispersion tells us how scattered data is around the mean.

Variance

- Average squared deviation from mean.
- Measures spread.
- Units are squared.

Standard Deviation

- Square root of variance.
- Same unit as original data.

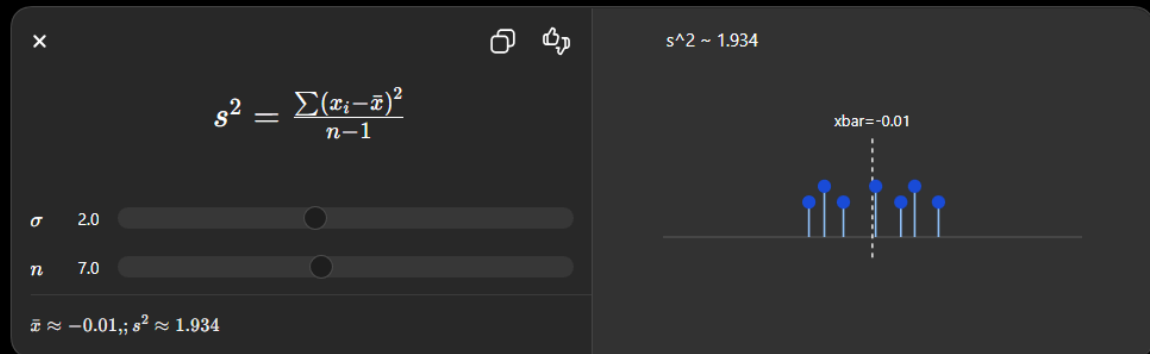
- Easier to interpret.

Key Formulas

Population Variance:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Sample Variance:



Standard Deviation:

$$\sigma = \sqrt{\sigma^2}$$

Rule to remember:

- Variance = average squared distance from mean.
- Standard Deviation = average spread of data around the mean in the original unit.

Distribution and Stationary Testing

What is the KS Test?

The K-S Test checks whether:

1. A sample follows a specified theoretical distribution.
2. Two samples come from the same distribution.

Since it does not require assumptions about population parameters, it is called a **non-parametric test**.

Main Idea

The test compares **cumulative frequencies**.

Instead of comparing means or variances, it compares:

Observed Cumulative Distribution

VS

Expected Cumulative Distribution

or

Sample 1 Distribution

VS

Sample 2 Distribution

The largest difference between the cumulative distributions becomes the test statistic.

KS Statistic

$$D = \max |F_o(x) - F_e(x)| \quad D = \max |F_o(x) - F_e(x)| \quad D = \max |F_o(x) - F_e(x)|$$

Where:

- $F_o(x)$ = Observed cumulative distribution
- $F_e(x)$ = Expected cumulative distribution
- D = KS statistic

The test finds the maximum absolute difference.

Types of KS Test

1. One-Sample KS Test

Used to check whether a sample follows a given distribution.

Example

Check whether student marks follow:

- Normal Distribution
- Uniform Distribution

Hypotheses

Null Hypothesis

H_0 : Sample follows the specified distribution

Alternative Hypothesis

H_1 : Sample does not follow the specified distribution

2. Two-Sample KS Test

Used to compare two independent samples.

Example

Compare:

- Sales before advertisement
- Sales after advertisement

Hypotheses

Null Hypothesis

H_0 : Both samples come from the same population

Alternative Hypothesis

H_1 : Samples come from different populations

Procedure

Step 1

Arrange observations in ascending order.

Example:

2, 4, 5, 8, 10

Step 2

Calculate cumulative frequencies.

X	Cumulative Frequency
2	1
4	2
5	3
8	4
10	5

Step 3

Convert cumulative frequencies into cumulative probabilities.

For sample size n :

$$F_o(x) = \frac{CF}{n}$$

Step 4

Find theoretical cumulative probabilities.

For example, from the Normal Distribution table.

Step 5

Calculate differences:

$$|F_o(x) - F_e(x)|$$

for every observation.

Step 6

Choose the maximum difference.

That maximum value becomes:

D_{cal}

(calculated KS statistic)

Step 7

Compare with critical table value:

D_{tab}

Decision rule:

If $D_{cal} > D_{tab}$

Reject H_0

If $D_{cal} \leq D_{tab}$

Accept H_0

Small Sample KS Critical Value

For one-sample tests, Dr. Harish Garg often presents:

$$D_{\alpha} = \frac{1.36}{\sqrt{n}} \quad D_{\alpha} = 1.36$$

for significance level:

$$\alpha = 0.05$$

where:

- n = sample size

This gives the critical value used in hypothesis testing.

Advantages of KS Test

No Distribution Assumption

Works even when data is not normal.

Simple to Apply

Uses cumulative frequencies.

Suitable for Small Samples

Can be applied to relatively small datasets.

Non-Parametric

No need to estimate mean and variance assumptions.

Limitations

Less Powerful Than Parametric Tests

When normality assumptions hold, parametric tests may perform better.

Sensitive Near Center

More sensitive around the middle of the distribution than at the tails.

Continuous Data Requirement

Works best for continuous distributions.

Typical University Exam Question

Q. What is the Kolmogorov–Smirnov Test?

Answer:

The Kolmogorov–Smirnov (K-S) Test is a non-parametric statistical test used to determine whether a sample follows a specified distribution or whether two samples come from the same distribution. It is based on the maximum absolute difference between cumulative distribution functions. If the calculated KS statistic exceeds the critical value, the null hypothesis is rejected.

Quick Revision Notes

KS Test

- Non-parametric test

- Based on cumulative distributions
- Uses maximum difference

Formula

$$D = \max |F_o(x) - F_e(x)| \quad D = \max |F_o(x) - F_e(x)| \quad D = \max |F_o(x) - F_e(x)|$$

Used For

- Testing normality
- Goodness of fit
- Comparing two samples

Decision Rule

$D_{cal} > D_{tab} \rightarrow \text{Reject } H_0$
 $D_{cal} \leq D_{tab} \rightarrow \text{Accept } H_0$

Important Feature

KS Test compares entire distributions, not just means or variances. This is why it is widely used in non-parametric hypothesis testing.

Model Monitoring Basics

1. What is Model Monitoring?

Model Monitoring is the continuous tracking of a machine learning model after deployment.

The goal is to ensure that the model remains:

- Accurate
- Reliable
- Stable
- Fair
- Relevant to current data

A deployed model is not a "train once and forget forever" system. Real-world environments continuously change.

2. Why Do Models Degrade?

When a model is trained, it assumes:

Training Data Distribution

≈

Future Production Data Distribution

But over time this assumption breaks.

Examples:

E-commerce

Customer buying behavior changes.

Banking

Loan applicant profiles evolve.

Healthcare

Disease patterns change.

Social Media

User interests shift rapidly.

When data changes significantly, model performance starts dropping. This phenomenon is called **drift**.

3. What is Data Drift?

Data Drift occurs when the statistical distribution of input features changes between training and production data.

Training:

Age Distribution

20–40 years

Production:

Age Distribution

40–70 years

The model now receives unfamiliar inputs.

This causes prediction instability.

Mathematical View

Suppose:

Training feature distribution:

$P_{\text{train}}(X)$

Production feature distribution:

$P_{\text{prod}}(X)$

Data drift occurs when:

$P_{\text{train}}(X) \neq P_{\text{prod}}(X)$ $P_{\text{train}}(X) = P_{\text{prod}}(X)$

4. Types of Drift

A. Data Drift (Covariate Drift)

Input features change.

Example:

Training:

Average Salary = ₹30,000

Production:

Average Salary = ₹70,000

Feature distributions have shifted.

B. Concept Drift

Relationship between input and output changes.

Example:

Before:

High Income → Low Default Risk

After recession:

High Income → Higher Default Risk

The underlying pattern itself changes.

C. Prediction Drift

Distribution of model predictions changes.

Example:

Earlier:

Fraud Probability
Mostly 0.1–0.3

Now:

Fraud Probability
Mostly 0.7–0.9

Prediction patterns have shifted.

5. What Should Be Monitored?

DataTrek emphasizes monitoring multiple layers.

A. Input Features

Monitor:

- Mean
- Median
- Variance

- Missing Values
- Feature Distributions

Example:

Age Mean

Training = 30

Production = 48

Possible drift signal.

B. Predictions

Monitor:

- Prediction distributions
- Confidence scores
- Probability scores

Example:

Normal Prediction Probability:

0.2–0.6

Current:

0.8–0.95

Potential issue.

C. Model Performance

Monitor:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Example:

Training Accuracy = 95%

Current Accuracy = 72%

Major warning sign.

D. System Metrics

Monitor:

- Latency
- Throughput
- CPU usage
- Memory usage

MLOps also includes infrastructure monitoring.

6. How is Drift Detected?

Several statistical methods are commonly used.

A. KS Test (Kolmogorov-Smirnov Test)

Measures difference between two distributions.

$$KS = \max |F_1(x) - F_2(x)|$$

Higher KS value:

Higher Distribution Difference

Used heavily in monitoring pipelines.

B. Population Stability Index (PSI)

Popular in banking and risk models.

Formula:

$$PSI = \sum (Actual\% - Expected\%) \ln \left(\frac{Actual\%}{Expected\%} \right)$$

Interpretation:

PSI	Meaning
< 0.1	No Drift
0.1–0.25	Moderate Drift
> 0.25	Significant Drift

C. Jensen-Shannon Distance

Measures similarity between probability distributions.

Frequently used in modern monitoring systems.

D. Wasserstein Distance

Measures how much one distribution must shift to become another.

Useful for continuous variables.

7. Real Production Workflow

Training Stage:

Train Model



Store Baseline Statistics



Deploy Model

Production Stage:

New Data Arrives



Compare With Baseline



Detect Drift



Generate Alerts



Retrain if Necessary

This is the standard MLOps monitoring lifecycle.

8. Alerting System

DataTrek discusses automatic alerts.

Example:

If $PSI > 0.25$

OR

Accuracy Drops Below 80%

Trigger:

- Email Alert
- Slack Alert
- PagerDuty Alert

Engineers investigate the issue.

9. What Happens After Drift Detection?

Several actions are possible.

Retraining

Train model on recent data.

Feature Engineering

Update feature pipelines.

Threshold Tuning

Adjust decision boundaries.

Replace Model

Deploy a better version.

Fallback Model

Switch to backup model if performance collapses.

10. Tools Used for Model Monitoring

Common tools mentioned across MLOps ecosystems include:

- Evidently AI
- MLflow
- Prometheus
- Grafana
- WhyLabs
- Fiddler AI

These tools track:

- Drift
 - Performance
 - Fairness
 - Data quality
 - Alerts
-

Interview Question

What is Data Drift?

Data Drift occurs when the statistical properties of production data change from the training data distribution. As a result, model predictions may become less reliable. Drift detection techniques such as KS Test, PSI, Jensen-Shannon Distance, and Wasserstein Distance are used to identify these changes and trigger retraining or corrective actions.

Quick Revision Notes

Model Monitoring

Tracks:

- Inputs
 - Predictions
 - Performance
 - Infrastructure
-

Data Drift

Input distribution changes.

$$P_{\text{train}}(X) \neq P_{\text{production}}(X) \quad P_{\text{train}}(X) = P_{\text{production}}(X)$$

Concept Drift

Relationship changes.

$$P(Y|X)_{\text{train}} \neq P(Y|X)_{\text{production}} \quad P(Y|X)_{\text{train}} = P(Y|X)_{\text{production}}$$

Prediction Drift

Prediction distribution changes.

Drift Detection Methods

- KS Test
 - PSI
 - Jensen-Shannon Distance
 - Wasserstein Distance
-

Production Response

Drift Detected



Alert Generated



Root Cause Analysis



Retrain / Replace Model

This monitoring loop is one of the most important responsibilities of an ML Engineer because a highly accurate model during training can become ineffective in production if drift is not detected and handled.

Normal Distribution and Its Empirical Formula

What is a Normal Distribution?

A **Normal Distribution** (also called a **Gaussian Distribution**) is one of the most important probability distributions in statistics, data science, and machine learning.

It describes data that is distributed symmetrically around the mean, where most observations cluster near the center and fewer observations occur farther away.

The graph looks like a **bell-shaped curve**.

Examples in Real Life

- Heights of people
- IQ scores
- Measurement errors
- Exam scores (often approximately)
- Blood pressure readings

Characteristics of a Normal Distribution

1. Bell-Shaped Curve

The distribution forms a symmetric bell shape.

2. Symmetric Around the Mean

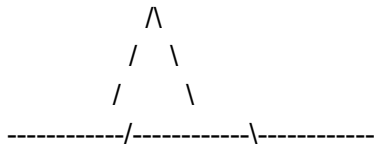
The left and right sides are mirror images.

3. Mean = Median = Mode

All three measures of central tendency are equal and located at the center.

Mean = Median = Mode





4. Total Area Under Curve = 1

The entire probability under the curve equals 100%.

Parameters of Normal Distribution

A normal distribution is completely defined by:

Mean (μ)

Controls the center of the distribution.

Standard Deviation (σ)

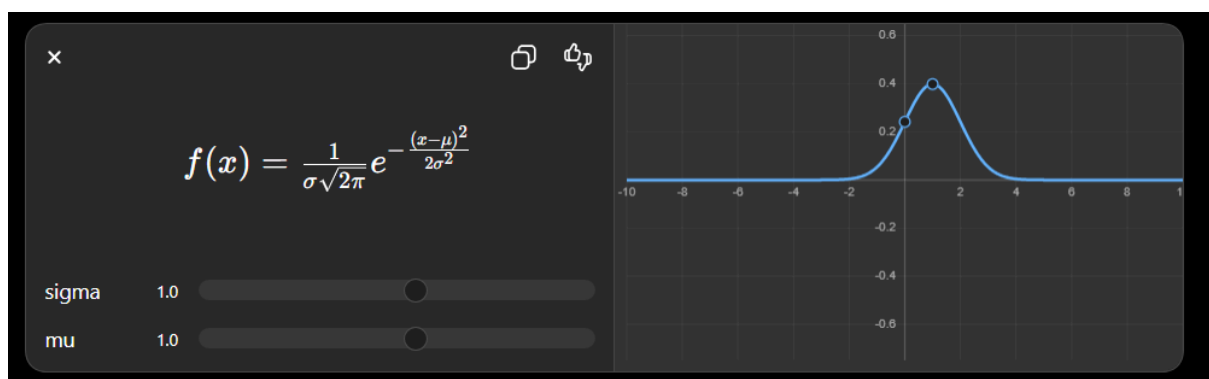
Controls the spread of the distribution.

A larger standard deviation produces a wider curve.

A smaller standard deviation produces a narrower curve.

Probability Density Function (PDF)

The mathematical equation of a normal distribution is:

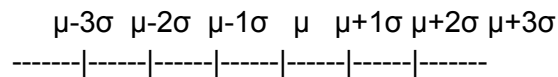


For interviews, you usually need to understand the intuition rather than memorize the formula.

The Empirical Rule (68–95–99.7 Rule)

This is the most important concept emphasized in introductory statistics.

The empirical rule tells us how data is distributed around the mean.



Within 1 Standard Deviation

Approximately:

68%68\%68%

of data lies between:

$\mu-\sigma$ and $\mu+\sigma$ $\mu-\sigma$ and $\mu+\sigma$

Within 2 Standard Deviations

Approximately:

95%95\%95%

of data lies between:

$\mu-2\sigma$ and $\mu+2\sigma$ $\mu-2\sigma$ and $\mu+2\sigma$

Within 3 Standard Deviations

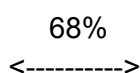
Approximately:

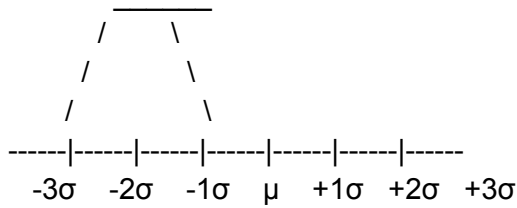
99.7%99.7\%99.7%

of data lies between:

$\mu-3\sigma$ and $\mu+3\sigma$ $\mu-3\sigma$ and $\mu+3\sigma$

Visual Representation





68% within $\pm 1\sigma$
95% within $\pm 2\sigma$
99.7% within $\pm 3\sigma$

Example

Suppose student marks have:

- Mean = 70
- Standard Deviation = 10

68% Rule

Marks lie between:

$$70 - 10 = 60$$
$$70 + 10 = 80$$

So approximately **68%** of students score between **60 and 80**.

95% Rule

Marks lie between:

$$70 - 20 = 50$$
$$70 + 20 = 90$$

So approximately **95%** of students score between **50 and 90**.

99.7% Rule

Marks lie between:

$$70 - 30 = 40$$
$$70 + 30 = 100$$

So approximately **99.7%** of students score between **40 and 100**.

Why is the Empirical Rule Important?

Outlier Detection

Values outside:

$$\mu \pm 3\sigma$$

are often considered unusual or potential outliers.

Data Analysis

Helps understand the spread of data quickly.

Machine Learning

Used in:

- Standardization
 - Feature Scaling
 - Anomaly Detection
 - Statistical Modeling
-

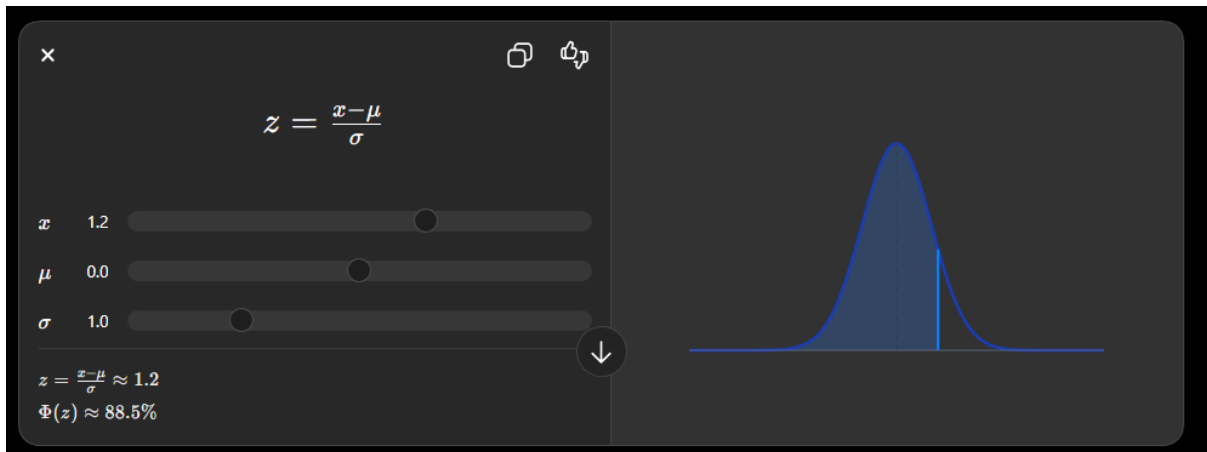
Standard Normal Distribution

A special normal distribution where:

$$\mu=0, \sigma=1$$

Any normal variable can be converted into a standard normal variable using the Z-score.

Z-Score Formula



Where:

- x = observation
- μ = mean
- σ = standard deviation

The Z-score tells us how many standard deviations a value is from the mean.

Interview Questions

Why is the Normal Distribution Important?

Many real-world phenomena approximately follow a normal distribution, and many statistical methods assume normality.

What is the Empirical Rule?

In a normal distribution, approximately 68% of observations lie within 1 standard deviation of the mean, 95% lie within 2 standard deviations, and 99.7% lie within 3 standard deviations.

Why is the Normal Distribution Called Bell-Shaped?

Because its graph is symmetric and resembles a bell.

Quick Revision Notes

Normal Distribution

- Bell-shaped
- Symmetric
- Mean = Median = Mode
- Area under curve = 1

Parameters

- Mean (μ)
- Standard Deviation (σ)

Empirical Rule

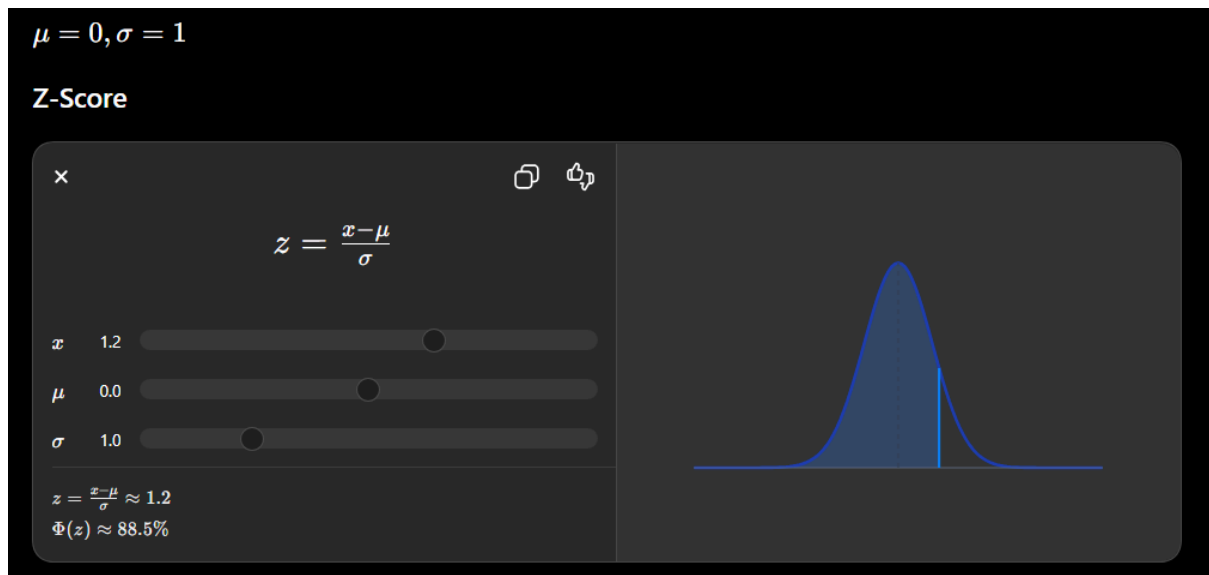
Range	Percentage of Data
-------	--------------------

$\mu \pm 1\sigma$	68%
-------------------	-----

$\mu \pm 2\sigma$	95%
-------------------	-----

$\mu \pm 3\sigma$	99.7%
-------------------	-------

Standard Normal Distribution



Key takeaway: The empirical rule (68–95–99.7) provides a quick way to understand how data is distributed around the mean in a normal distribution and is one of the most frequently asked concepts in statistics and

Central Limit Theorem (CLT) intuition

The **Central Limit Theorem** is one of the most beautiful ideas in probability.

1. The core idea

The Central Limit Theorem says:

If you repeatedly take **samples** from any distribution and compute their **average**, the distribution of those averages becomes approximately **normal (bell-shaped)** — even if the original data is not normal.

In simple words

Even if your original data looks weird, skewed, or uneven...

👉 the *averages of many samples* will look **normal**.

2. Intuition

Imagine this process:

Step 1: Start with any distribution

It could be:

- Uniform (flat)
- Skewed
- Weird-shaped

Example:

Original distribution (anything):



No bell shape required.

Step 2: Take random samples

Pick:

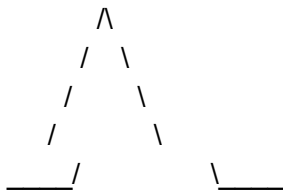
- 5 numbers → compute mean
- again 5 numbers → compute mean
- again → repeat thousands of times

Now you are collecting **sample means**, not raw data.

Step 3: Plot those means

Something surprising happens:

Distribution of sample means:



A normal distribution appears out of nowhere.

3. Why does this happen? (Intuition)

Think like 3Blue1Brown visualization:

Every sample mean is a “sum of random effects”

A mean is just:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

So you're adding many random variables together.

Key insight:

Addition smooths randomness

- Extreme values cancel out
- Small fluctuations average out

- Many independent effects combine

So the result becomes **stable and symmetric**

4. The “shape-building” intuition

Think of each random variable as a “shape contributor”.

When you:

- add 2 variables → shape becomes smoother
- add 10 variables → even smoother
- add 100 variables → bell shape appears

This is why normal distribution is called a “**sum-of-random-effects attractor**”.

5. Important conditions (CLT assumptions)

The theorem works when:

1. Independence

Samples should not strongly depend on each other.

2. Identical distribution (roughly)

They come from the same process.

3. Finite mean and variance

Not infinite extreme distributions.

6. What CLT does NOT say

- ✗ It does NOT say original data becomes normal
- ✗ It does NOT say single observations become normal

It only says:

The **distribution of sample means** becomes normal.

7. Mathematical version

If:

- $X_1, X_2, \dots, X_n$ are i.i.d random variables
- with mean μ and variance σ^2

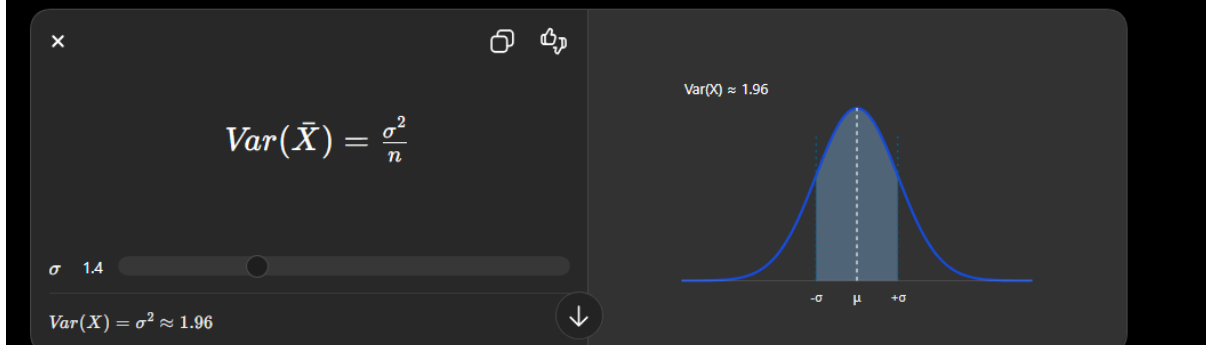
Then:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1)$$

As $n \rightarrow \infty$

8. Why variance shrinks

As sample size increases:



So:

- Bigger sample $\rightarrow$ less fluctuation
 - More stability $\rightarrow$ tighter bell curve
-

9. Simple real-life intuition

Example: Dice rolls

A single dice roll is uniform:

1 2 3 4 5 6 (equal probability)

Now:

- take 10 rolls → compute average
- repeat many times

Result:

👉 averages form a **normal distribution**

Even though dice is NOT normal.

10. Why CLT is powerful (3Blue1Brown perspective)

CLT explains why the normal distribution appears everywhere:

- heights
- measurement errors
- exam scores
- noise in signals
- stock returns (approx.)

Because many real-world values are:

sums of many small random influences

11. Machine Learning importance

CLT is the backbone of:

- Confidence intervals
 - Hypothesis testing
 - A/B testing
 - Error estimation
 - Sampling theory
-

12. One-line interview answer

The Central Limit Theorem states that the distribution of sample means of independent random variables approaches a normal distribution as sample size increases, regardless of the original distribution, provided the variables have finite mean and variance.